

1. Administrative & Study Identification

Full Study Title: A Phase 3 Randomized, Placebo-Controlled Clinical Study to Evaluate the Efficacy and Safety of MK-0616 in Reducing Major Adverse Cardiovascular Events in Participants at High Cardiovascular Risk **Short Title/Acronym:** CORALreef Outcomes (TIMI 77 / MK-0616-015) **Protocol Number:** MK-0616-015 **NCT Number:** NCT06008756 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Enlicitide decanoate (MK-0616), an oral PCSK9 inhibitor **Phase of Development:** Phase 3 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of enlicitide decanoate compared with placebo in increasing the time to the first occurrence of major adverse cardiovascular events (MACE), including coronary heart disease (CHD) death, ischemic stroke, myocardial infarction (MI), acute limb ischemia or major amputation, or urgent arterial revascularization. **Secondary Objectives:** To evaluate the efficacy and safety of enlicitide decanoate in lowering LDL-C and other lipid parameters (e.g., non-HDL-C, ApoB), as well as to assess overall safety and tolerability.

2.2 Background & Rationale Major cardiovascular events are the leading cause of global death, with high levels of low-density lipoprotein cholesterol (LDL-C) significantly contributing to this risk. While injectable PCSK9 inhibitors are available, MK-0616 offers a novel, once-daily oral formulation. This approach aims to provide a highly effective, more accessible treatment option to help a broader population of high-risk patients achieve their LDL-C targets and reduce the likelihood of MACE.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 14,550 participants **Number of Sites:** Global (approximately 674 locations) **Estimated Study Start:** 09-Oct-2023 **Estimated Primary Completion:** 29-Nov-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a history of a major atherosclerotic cardiovascular disease (ASCVD) event, or age ≥ 50 years at high risk for a first major ASCVD event. 2. Fasted lipid values of LDL-C ≥ 70 mg/dL (for history of ASCVD event) OR LDL-C ≥ 90 mg/dL (for high risk of first ASCVD event). 3. Currently taking a stable dose of background lipid-lowering therapies, including at least a moderate- or high-intensity statin, for at least 30 days prior to screening.

4.2 Exclusion Criteria 1. History of homozygous familial hypercholesterolemia (FH). 2. Severe heart failure (New York Heart Association Class IV), Left Ventricular Ejection Fraction $\leq 25\%$, or recent hospitalization for heart failure within 3 months. 3. Active or recent treatment with another PCSK9 inhibitor.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Once-daily tablet. Participants must fast for 8 hours before and 30 minutes after taking the treatment. **Route of Administration:** Oral **Duration of Treatment:** Up to approximately 6 years (average 4.5 to 6 years).

5.2 Concomitant & Prohibited Therapy Permitted: Background stable lipid-lowering therapies (e.g., statins, ezetimibe). **Prohibited:** Other PCSK9 inhibitors, angiopoietin-like protein 3 inhibitors, or microsomal triglyceride transfer protein inhibitors.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -30 to 0 Informed Consent, Medical History, Fasting Lipid Labs, ECG Baseline Day 0 Randomization, First Dose Interim Periodic Safety Labs, MACE Evaluation, Lipid Monitoring, Physical Exams End of Study Year 6 Final Vital Signs, Exit Interview, Investigational Product Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time to the first occurrence of major adverse cardiovascular events (MACE). **Measurement Method:** Clinical event adjudication by an independent committee tracking occurrences of CHD death, ischemic stroke, MI, acute limb ischemia/major amputation, or urgent arterial revascularization.

7.2 Secondary & Exploratory Endpoints - Percent change in LDL-C from baseline to target assessment dates. - Percent change in non-HDL-C, apolipoprotein B (ApoB), and lipoprotein(a) (Lp(a)).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical visits, routine ECGs, and laboratory tests. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee established to oversee trial safety and event adjudication.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for time-to-event primary efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N = 14,550$ participants powered to detect a significant increase in the time to first MACE event between the enlicotide decanoate and placebo groups. **Handling of Missing Data:** Standard censoring rules for time-to-event variables; multiple imputation models applied to continuous laboratory lipid parameter changes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine frequency of onsite and remote monitoring visits to ensure medication adherence and safety follow-up. **Audit Trail:** Secure eCRF locking and query resolution mechanisms applied per regulatory standards.

11. Study Identifiers & Governance

Primary ID: MK-0616-015 **Registry Number:** NCT06008756 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** CORALreef Outcomes **Official Regulatory Title:** A Phase 3 Randomized, Placebo-Controlled Clinical Study to Evaluate the Efficacy and Safety of MK-0616 in Reducing Major Adverse Cardiovascular Events in Participants at High Cardiovascular Risk

12. Clinical Framework (Cardiovascular Specific)

Primary Condition: High Cardiovascular Risk / Hypercholesterolemia **Diagnosis Threshold:** LDL-C ≥ 70 mg/dL (secondary prevention) OR ≥ 90 mg/dL (primary prevention)

Medical Methodology: Stable background lipid-lowering therapy (statin-based) **Optimization Target:** PCSK9-targeted LDL-C reduction to prevent MACE

13. Study Procedures & Methodology

Management Pathway: Oral PCSK9 inhibitor administration atop background standard of care **Education Protocol (HCP):** Trial protocol training for standardized MACE adjudication **Education Protocol (Patient):** Strict medication fasting protocol (8 hours pre-dose and 30 mins post-dose) **Quality Audit Mechanism:** Adherence tracking and long-term event surveillance

14. Observation & Follow-Up Schedule

Total Duration: Up to 6 years (Estimated completion 29-Nov-2029) **Onsite Visit Cadence:** Periodic standard visits per protocol schedule **Remote Monitoring Frequency:** Routine continuous tracking of clinical outcomes **Checklist Review Interval:** Continuous MACE and AE surveillance

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				